

ITCS223 Introduction to Web Development

Project Phase II

One Bank Kheow Store

Submitted by

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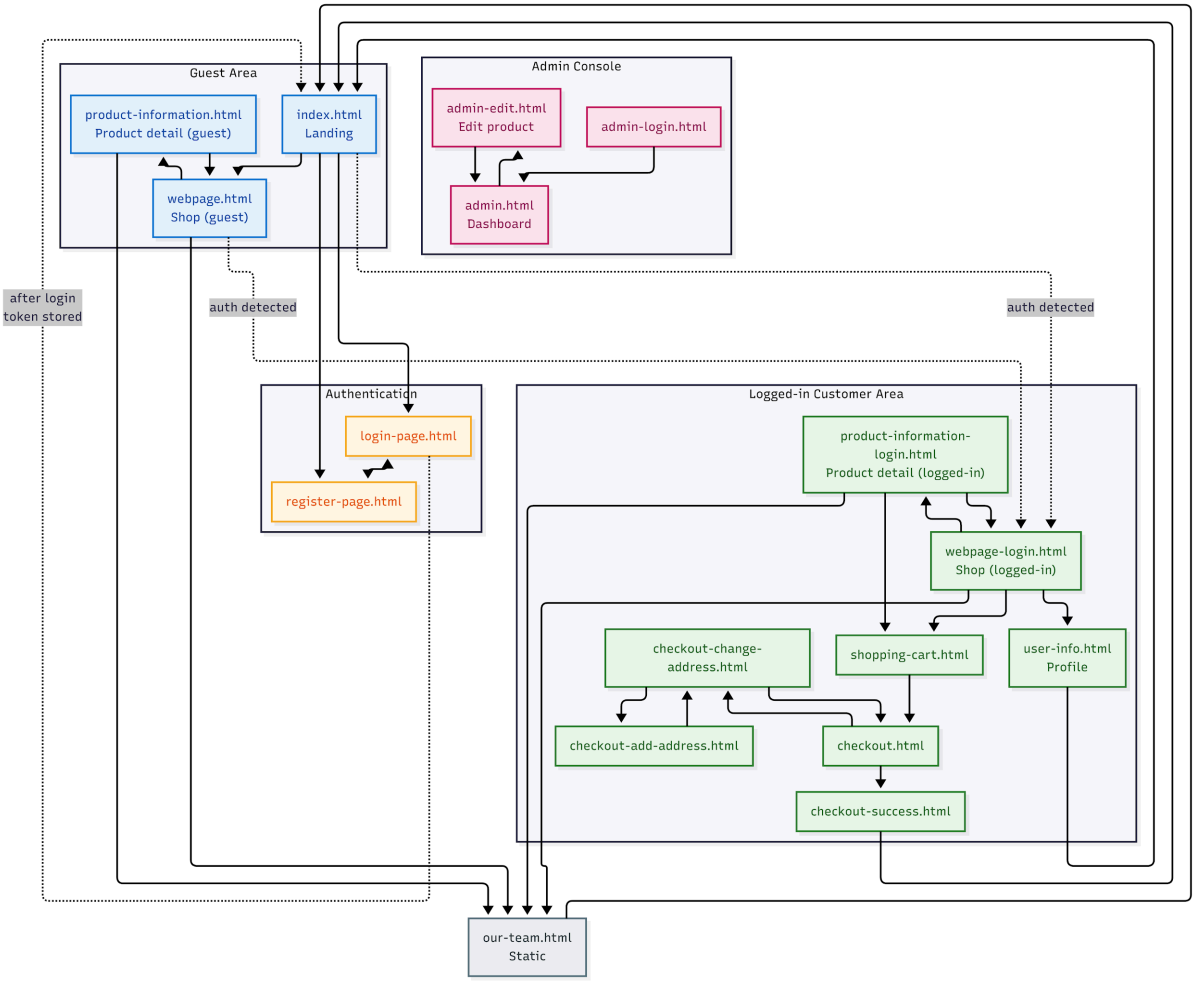
Business Overview

Our platform is based on the popular "20 Baht Shop" model, which is well-known for its fixed- price concept where all items sell for 20 THB. This concept is popular due to its affordability and accessibility within local communities. Traditionally, these shops operate as small physical stores, offering a wide range of affordable things, including household goods, stationery, kitchenware, toys, and daily necessities.

For that reason, we transformed this concept into a digital store. The website is centralized, and scalable to maintain the unique fixed-price idea. It includes key functionalities such as product management, customer registration, and comprehensive order processing. The system provides various payment methods, and delivery. The administrators have full authority to manage the product stock, while customers can easily browse products, place orders, and complete payments online.

Navigation Diagram

The application is divided into four navigation tiers: Public-Guest area, Authentication area, Logged-in Customer area(which includes the checkout funnel and the user profile), and an Admin Console. The static OurTeam page sits outside any tier and is reachable from every shop page on the footer.



Web App Detail and Code Implementation

1. index.html
Endpoint(s): GET /products/search (used to grab a few items for the recommended row),
GET /products/exchange-rate

- Public landing page. Shows hero banner, category icons, and a Recommended row of products with THB→CNY price annotation.

Code Explanation:

The page includes the shared script js/products.js, which calls loadProducts() on load:

```javascript
async function loadProducts(filters = {}) {
    const grid = document.querySelector(".product-grid");
    if (!grid) return;
    const [items, rateData] = await Promise.all([
        searchByNameOrId(filters),
        OBK.getExchangeRate(),
    ]);
    // …render cards
}
```

- searchByNameOrId() calls OBK.apiFetch("/products/search"), then OBK.getExchangeRate() hits /products/exchange-rate so each card can show its CNY equivalent.

2. webpage.html
Endpoint(s): GET /products/search

- Guest product listing / shop page. Renders the full catalogue and accepts filters from the navbar search.

Code Explanation:

In js/products.js, searchByNameOrId(filters) builds the query string and calls the API:

```javascript
async function searchByNameOrId(filters) {
    const qs = new URLSearchParams();
    if (filters.name)     qs.set("name", filters.name);
    if (filters.brand)    qs.set("brand", filters.brand);
    if (filters.category) qs.set("category", filters.category);
    const res = await OBK.apiFetch("/products/search?" + qs.toString());
    return res.data || [];
}
```

- The search is triggered by js/nav-search.js → applySearch(), which on a non-shop page redirects to webpage.html?name=…&brand=…&category=… instead.

3. product-information.html
Endpoint(s): GET /products/:productID, GET /products/search (for the Recommend row)

- Guest product detail page. Reads ?id= from the URL and shows full information (name, brand, description, image, price, stock).

Code Explanation:

In js/product-detail.js:

```JavaScript
async function loadProduct() {
    const id = getIdFromUrl();
    if (!id) { showError("No product ID in URL."); return; }
    const [res, rateData] = await Promise.all([
        OBK.apiFetch(`/products/${encodeURIComponent(id)}`),
        OBK.getExchangeRate(),
    ]);
    const p = res.data;
    document.querySelector(".product-title").textContent = p.productName;
    // …populate brand, description, image, stock, 20฿ + CNY
}
```

- loadRecommended() then calls /products/search, filters out the current product, shuffles, and slices the first 10.

4. login-page.html
Endpoint(s): POST /auth/customer/login

- Customer login form. On success, stores the customer token + ID in localStorage and redirects to webpage-login.html.

Code Explanation:

In js/login.js, the form submit handler is wired as:

```JavaScript
form.addEventListener("submit", async (e) => {
    e.preventDefault();
    const cEmail    = document.getElementById("email").value.trim();
    const cPassword = document.getElementById("password").value;
    const data = await OBK.apiFetch("/auth/customer/login", {
        method: "POST",
        body: { cEmail, cPassword },
```

```
    });
    localStorage.setItem("customerToken", data.token);
    localStorage.setItem("customerID",    data.cID);
    localStorage.setItem("customerEmail", data.cEmail);
    window.location.href = "webpage-login.html";
});
```

- The token (customer-secret) is what OBK.apiFetch later attaches as the Authorization header for any auth: "customer" request.

5. register-page.html
Endpoint(s): POST /auth/customer/login (re-uses the login flow after a client-side registration form)

- Customer registration form. Currently the form posts the new account info via the same OBK.apiFetch wrapper and then logs the user in.

Code Explanation:

js/login.js is reused the same module wires the right-hand side register form, performing field validation and then redirecting to login-page.html on success:

```
JavaScript
const data = await OBK.apiFetch("/auth/customer/login", {
    method: "POST",
    body: { cEmail, cPassword },
});
window.location.href = "login-page.html";
```

6. webpage-login.htmlw
Endpoint(s): GET /products/search
Description: Logged-in twin of webpage.html. Identical layout, but the navbar Web Component renders the customer cart/profile/logout instead of the Login button, and product cards expose Add to cart actions.
Code Explanation:

The same loadProducts() from js/products.js runs here. The only difference is the navbar — components/components.js reads localStorage.customerToken and renders accordingly:

```
JavaScript
const navIconsHtml = isCustomer
    ? `<a href="shopping-cart.html"><img …/></a>
       <a href="user-info.html"><img class="nav-profile-img" …/></a>
       <a href="#" id="nav-logout-btn">Logout</a>`
    : `<a href="login-page.html">Login / Register</a>`;
```

7. product-information-login.html

Endpoint(s): GET /products/:productID

- Logged-in twin of product-information.html. Adds the "Add to cart" button which writes the selected product into localStorage.cart.

Code Explanation:

Same loadProduct() from js/product-detail.js. The cart-add handler is local to this page:

```javascript
document.querySelector(".add-to-cart").addEventListener("click", () => {
    const cart = JSON.parse(localStorage.getItem("cart") || "[]");
    cart.push({ productID: p.productID, qty: 1 });
    localStorage.setItem("cart", JSON.stringify(cart));
});
```

8. shopping-cart.html

Endpoint(s): GET /products/exchange-rate, GET /products/:productID (per cart line, to fetch up-to-date stock/price)

- Cart review page. Reads localStorage.cart, fetches each product's current details, and shows totals with a promo-code input.

Code Explanation:

Inside js/products.js cart routine:

```javascript
const cart = JSON.parse(localStorage.getItem("cart") || "[]");
const items = await Promise.all(
    cart.map((line) =>
        OBK.apiFetch(`/products/${encodeURIComponent(line.productID)}`)
           .then((r) => ({ ...r.data, qty: line.qty }))
    )
);
const subtotal = items.reduce((s, it) => s + 20 * it.qty, 0);
```

js/exchange-rate.js → applyExchangeRate() then runs across .item-price, .item-total, and .summary-row .value to inject the (~ X.XX ¥) annotation.

9. checkout.html

Endpoint(s): GET /user/profile/:cID (to read the saved address)

- Final checkout view. Shows the chosen delivery address, the cart line items, and lets the customer pick a payment method (PromptPay / Credit Card / TrueMoney).

Code Explanation:

```
JavaScript
const cID = localStorage.getItem("customerID");
const profile = await OBK.apiFetch(
    `/user/profile/${encodeURIComponent(cID)}`,
    { auth: "customer" }
);
document.querySelector(".address-text").textContent = profile.data.cAddress;
```

- The "Change address" button links to checkout-change-address.html; the "Confirm purchase" button clears the cart and redirects to checkout-success.html.

10. checkout-change-address.html

Endpoint(s): GET /user/profile/:cID

- Let the customer pick which saved address to use for delivery (or jump to add a new one).

Code Explanation:

js/user-info.js (re-used here) loads the current saved address:

```
JavaScript
const res = await OBK.apiFetch(
    `/user/profile/${encodeURIComponent(session.cID)}`,
    { auth: "customer" }
);
document.querySelector("[name='address']").value = res.data.cAddress;
```

- The "Add new" button navigates to checkout-add-address.html; "Confirm" returns to checkout.html.

11. checkout-add-address.html
Endpoint(s): PUT /user/profile/:cID

- Form to write a new delivery address into the customer profile.

Code Explanation:

In js/user-info.js, the same submit handler is reused:

```
JavaScript
await OBK.apiFetch(
    `/user/profile/${encodeURIComponent(session.cID)}`,
    {
        method: "PUT",
        auth: "customer",
        body: { cAddress: addressInput.value.trim(), /* …other required fields */
},
    }
);
window.location.href = "checkout-change-address.html";
```

- The endpoint requires every editable profile field, so the handler re-sends the existing values for fields it isn't changing.

12. checkout-success.html
Endpoint(s): (none)

- Static confirmation page shown after a successful purchase. Clears localStorage.cart on load and offers a "Continue shopping" link back to index.html.

Code Explanation:

```
JavaScript
<script>
    localStorage.removeItem("cart");
</script>
<a href="index.html" class="back-btn">Continue shopping</a>
```

13. user-info.html

Endpoint(s): GET /user/profile/:cID, PUT /user/profile/:cID

- Customer profile page. Shows current details and lets the user update first/last name, phone, DOB, gender, address, and profile picture URL.

Code Explanation:

In js/user-info.js:

```JavaScript
async function init() {
    const session = OBK.requireCustomer();          // redirect if not logged in
    const res = await OBK.apiFetch(
        `/user/profile/${encodeURIComponent(session.cID)}`,
        { auth: "customer" }
    );
    populateForm(res.data);
}

form.addEventListener("submit", async (e) => {
    e.preventDefault();
    await OBK.apiFetch(
        `/user/profile/${encodeURIComponent(session.cID)}`,
        {
            method: "PUT",
            auth: "customer",
            body: {
                phoneNo:    phoneInput.value.trim(),
                cAddress:   addressInput.value.trim(),
                cFirstName: firstName.value.trim(),
                cLastName:  lastName.value.trim(),
                cGender:    genderInput.value,
                cBirthDate: dobInput.value || null,
                cPicture:   pictureInput.value.trim(),
            },
        }
    );
});
```

- OBK.requireCustomer() guards the page — if no customerToken is in localStorage, the user is redirected to login-page.html.

14. admin-login.html
Endpoint(s): POST /auth/admin/login

- Admin authentication page. Stores the admin token in localStorage.adminToken on success and redirects to the dashboard.

Code Explanation:

In js/admin-login.js:

```javascript
form.addEventListener("submit", async (e) => {
    e.preventDefault();
    const userName = document.getElementById("Username").value.trim();
    const password = document.getElementById("password").value;
    const data = await OBK.apiFetch("/auth/admin/login", {
        method: "POST",
        body: { userName, password },
    });
    localStorage.setItem("adminToken", data.token);   // = 'supersecret'
    window.location.href = "admin.html";
});
```

- The token is then attached as Authorization: supersecret for every admin product call.

15. admin.html
Endpoint(s): GET /products/admin, DELETE /products/admin/:productID

- Admin dashboard listing every product with edit / delete buttons and an Add Item shortcut to admin-edit.html.

Code Explanation:

In js/admin-products.js:

```javascript
async function load() {
    OBK.requireAdmin();
    const res = await OBK.apiFetch("/products/admin", { auth: "admin" });
    renderList(res.data);
}

function wireDeleteDelegation() {
    document.addEventListener("click", async (e) => {
        const btn = e.target.closest(".btn-delete");
        if (!btn) return;
        const id = btn.getAttribute("data-product-id");
        if (!confirm(`Delete product ${id}?`)) return;
```

```
        await OBK.apiFetch(`/products/admin/${encodeURIComponent(id)}`, {
            method: "DELETE",
            auth: "admin",
        });
        load();   // refresh list
    });
}
```

16. admin-edit.html

Endpoint(s): GET /products/admin/:productID, POST /products/admin, PUT /products/admin/:productID

- Single form used both to create a new product (no ?id= in URL) and to edit an existing one (?id=Pxxxxx).

Code Explanation:

In js/admin-edit.js:

JavaScript

```
async function init() {
    OBK.requireAdmin();
    const id = new URLSearchParams(location.search).get("id");
    if (id) {                                   // edit mode → preload
        const res = await OBK.apiFetch(
            `/products/admin/${encodeURIComponent(id)}`,
            { auth: "admin" }
        );
        fillForm(res.data);
    }

    form.addEventListener("submit", async (e) => {
        e.preventDefault();
        const body = collectForm();
        if (id) {
            await OBK.apiFetch(`/products/admin/${encodeURIComponent(id)}`,
                { method: "PUT",  auth: "admin", body });
        } else {
            await OBK.apiFetch("/products/admin",
                { method: "POST", auth: "admin", body });
        }
        window.location.href = "admin.html";
    });
}
```

17. our-team.html
Endpoint(s): (none)

- Static "Our Team" showcase reachable from every shop page footer. Pure HTML/CSS no JavaScript or backend calls.
- No code logic; the page only contains team member cards and the shared <nav-bar> and footer markup.

Web Service Detail and Code Implementation

1. Product Search Service
Endpoint(s): GET /products/search

The service supports four optional criteria:

- No criteria search
- Name
- Brand
- Category

Code Explanation:

In productRoutes.js, the route is defined as:

```
JavaScript
router.get('/search', productController.searchProducts);
```

The request is sent to searchProducts() inside productController.js, where query parameters are collected:

```
JavaScript
const filters = {
    name: req.query.name,
    brand: req.query.brand,
    category: req.query.category,
};
```

Then the controller calls Product.searchProducts() from productModel.js

Inside the model, dynamic SQL is created using:

```
SQL
SELECT * FROM Product WHERE 1=1
```

2. Product Detail Service
Endpoint(s): GET /products/:productID

- This service allows users to view complete information of a specific product using its product ID.

Code Explanation:

The route is defined as:

```JavaScript
router.get('/:productID', productController.getProductDetail);
```

The controller extracts the productID from URL parameters:

```JavaScript
const { productID } = req.params;
```

Then it calls:

```JavaScript
Product.getProductById(productID)
```

The model executes:

```SQL
SELECT * FROM Product WHERE productID = ?
```

3. Recommended Products Service

Endpoint(s): GET /products/recommend

- This service randomly selects products from the database to display recommended products to users.

Code Explanation:

Example: http://localhost:3100/products/recommend?limit=5

The system accepts an optional limit parameter. If not provided, the default value is 10.

The SQL query used is:

```SQL
SELECT * FROM Product ORDER BY RAND() LIMIT ?
```

4. Admin Product Management

- This service allows administrators to manage product records:

Endpoint(s): GET /products/admin
GET /products/admin/:productID
POST /products/admin
PUT /products/admin/:productID
DELETE /products/admin/:productID

Related Codes and explanations:

4.1 In productController.js:

```
JavaScript
router.get('/admin', verifyToken, productController.getAllProducts);
router.get('/admin/:productID', verifyToken, productController.getProductById);
router.post('/admin', verifyToken, productController.createProduct);
router.put('/admin/:productID', verifyToken, productController.updateProduct);
router.delete('/admin/:productID', verifyToken, productController.deleteProduct);
```

These routes handle:

1. View all products
2. View specific products
3. Insert products
4. Update products
5. Delete products

And all require Authorization: supersecret

4.2 In productController.js:

```
JavaScript
getAllProducts()
getProductById()
createProduct()
updateProduct()
deleteProduct()
```

- These are the functions to manage product records.

4.3 In productModel.js, it contains the SQL queries, for example, Insert, Update, and Delete, to query to the database directly.

4.4 In authMiddleware.js, it protects admin routes.

- It checks if Authorization: supersecret exists in the request header or not.

5. Admin Authentication

Endpoint(s): POST /auth/admin/login

When an admin submits login information, the system checks the entered credentials against the data stored in the AdminLoginInfo table in the database. If the login is successful, the system returns a verification token that grants the administrator permission to access protected controller functions, such as product management (insert, update, and delete). If the username or password is incorrect, the system returns an authentication error message and denies access.

- This service allows the system to check if the input username and password of the admin are valid.

Code Explanation:

In authRoutes.js, the route is defined as:

```JavaScript
router.post('/admin/login', authController.adminLogin);
```

The controller:

1. Receives userName and password
2. Validates that both fields are provided.
3. Calls Auth.findAdmin() from authModel.js

Then the model executes:

```SQL
SELECT * FROM AdminLoginInfo
WHERE userName = ? AND password = ?
```

If the admin exists:

- login time is updated using Log.updateLastLogin()
- token supersecret is returned

In authMiddleware.js, it checks if the token matches Authorization: supersecret before permitting any access.

6. Customer Authentication

Endpoint(s): POST /auth/customer/login

- This service allows the system to check if the input username and password of the customer are valid before accessing related functions such as viewing and updating profile information.

Code Explanation:

In authRoutes.js, the route is defined as:

```JavaScript
router.post('/customer/login', authController.customerLogin);
```

The request is handled by customerLogin() from authController.js

The controller:

1. Receives cEmail and cPassword
2. Validates required fields.
3. Calls Auth.findCustomer() from authModel.js

Then the model executes:

```SQL
SELECT cID, cEmail, cPassword
FROM Customer
WHERE cEmail = ? AND cPassword = ?
```

If login is successful:

- token customer-secret is returned
- customer ID and email are returned

In authMiddleware.js, it checks if the token matches Authorization: customer-secret before permitting any access.

7. User Profile Service
Endpoint(s): GET /user/profile/:cID
PUT /user/profile/:cID

- This service allows customers to view and update personal profile information.

Related codes and explanations:
In userRoutes.js (defines endpoints):

```JavaScript
router.get('/profile/:cID', verifyToken, userController.getProfile);
router.put('/profile/:cID', verifyToken, userController.updateProfile);
```

These routes handle:
- View customer profile
- Update customer profile

In userController.js (logic):

```JavaScript
getUserProfile(cID, (err, results) => { ... };
```

- The controller receives the customer ID from the URL, calls the model, and returns the result as JSON.

```JavaScript
updateUserProfile(cID, updateData, (err, results) => { ... };
```

- It validates input data and sends updated values to the database.

In userModel.js, it contains SQL queries, for examples:
Get profile

```SQL
SELECT * FROM Customer WHERE cID = ?
```

Update profile

```SQL
UPDATE Customer
SET phoneNo = ?, cAddress = ?, cFirstName = ?, cLastName = ?, cGender = ?,
cBirthDate = ?, cPicture = ?
WHERE cID = ?
```

- They directly communicate with the Customer table.

8. Exchange Rate Service

Endpoint(s): GET /products/exchange-rate

- This service includes a public web service (https://www.exchangerate-api.com/) also, allowing the system to fetch and calculate real-time currency exchange rates. It retrieves the conversion rate from Thai Bath (THB) to Chinese Yuan (CNY) and calculates a sample converted price.

Related codes and explanations:

In productRoutes.js (defines endpoints):

```
JavaScript
router.get('/exchange-rate', productController.getExchangeRate);
```

In productController.js:

```
JavaScript
const axios = require('axios');
```

- Imports axios for external API request

```
JavaScript
const getExchangeRate = async (req, res) => {
    try {
        const apiKey = process.env.EXCHANGE_RATE_KEY;
        const url = `https://v6.exchangerate-api.com/v6/${apiKey}/latest/THB`;

        const response = await axios.get(url);
        const cnyRate = response.data.conversion_rates.CNY;
        const priceInCNY = (20 * cnyRate).toFixed(2);

        res.json({
            error: 0,
            base: 'THB',
            target: 'CNY',
            rate: cnyRate,
            priceTHB: 20,
            priceCNY: priceInCNY
        });
    } catch (err) {
        console.error("Exchange Rate API Error:", err.message);
        res.status(500).json({ error: 1, message: "Could not fetch exchange rate"
});
```

```
    }
};
```

- The endpoint integrates with the ExchangeRate-API (v6) to get the latest fiat currency exchange rates using THB as the base currency.
- After receiving the response, the system calculates the equivalent value of a fixed product price (20 THB) in CNY and returns the result as JSON.
- The controller uses async/await to handle the asynchronous HTTP GET request to the external API URL: https://v6.exchangerate-api.com/v6/{API_KEY}/latest/THB

```
JavaScript
# API Key
EXCHANGE_RATE_KEY=beb67ecfdca1092a2d7d4eb7
```

- The API key is stored in a .env file to securely use it (process.env.EXCHANGE_RATE_KEY), preventing exposed sensitive credentials.

Test Results (e.g. Postman)

Test Case 1: Search with no criteria
Method: Get
URL: http://localhost:3100/products/search
Result:


```
{
  "error": 0,
  "count": 20,
  "data": [
    {
      "productID": "P00001",
      "adminID": "A001",
      "productName": "Multi-color Plastic Basket",
      "productAmount": 500,
      "productCategory": "Kitchen",
      "productImage": "https://www.bakerross.co.uk/media/catalog/product/f/c/fc892v-c2.jpg",
      "productBrand": "-",
      "productDetail": "Versatile multi-purpose plastic storage basket. Made from high-quality Grade A PP plastic, durable and sturdy. Perfect for organizing office desks, bathrooms, or kitchen cabinets. Breathable design prevents moisture buildup."
    },
    {
      "productID": "P00002",
      "adminID": "A002",
      "productName": "Pack of 3 Blue Ballpoint Pens",
      "productAmount": 300,
      "productCategory": "Stationery",
      "productImage": "https://pim-cdn0.ofm.co.th/products/original/1891584.jpg?v=20260330&x-image-process=image/format,webp",
      "productBrand": "Double A",
      "productDetail": "Set of 3 premium blue ink ballpoint pens with a 0.6mm needle tip. Provides a smooth, skip-free writing experience. Quick-drying ink prevents smudging, making it ideal for students and office professionals."
    },
    {
      "productID": "P00003",
      "adminID": "A003",
      "productName": "Stainless Steel Vegetable Peeler",
      "productAmount": 150,
      "productCategory": "Kitchen",
      "productImage": "https://www.vipshopboutic.com/cdn/shop/products/178172_2.jpg?v=1692863993&width=1445",
      "productBrand": "Zebra",
      "productDetail": "High-quality fruit and vegetable peeler featuring a sharp stainless steel blade. Rust-resistant and long-lasting. Ergonomically designed handle provides a comfortable grip, making food preparation effortless and safe."
    },
    {
      "productID": "P00004",
      "adminID": "A004",
      "productName": "Microfiber Cleaning Cloth",
      "productAmount": 400,
      "productCategory": "Cleaning",
      "productImage": "https://www.nitori.co.th/cdn/shop/files/898245132_640x640.jpg?v=1739701389",
      "productBrand": "NITORI",
      "productDetail": "Ultra-soft microfiber cloth with 6x higher water absorption than standard towels. Scratch-free and lint-free. Perfect for cleaning delicate surfaces like glassware, dinnerware, or electronic screens."
    },
    {
      "productID": "P00005",
      "adminID": "A005",
      "productName": "Crispy Fish Snack Pack",
      "productAmount": 200,
      "productCategory": "Snacks",
      "productImage": "https://st.bigc-cs.com/cdn-cgi/image/format=webp,quality=90/public/media/catalog/product/14/88/8852044592414/8852044592414_1-20230323160759-.jpg",
      "productBrand": "Taro",
      "productDetail": "Crispy baked fish snacks with a spicy and bold flavor. Low calorie, high protein, and no preservatives. Convenient portable size, perfect for a mid-afternoon snack or enjoying during a movie."
    },
```

```
    {
        "productID": "P00006",
        "adminID": "A004",
        "productName": "Gaiatop Mini Desk Fan (USB)",
        "productAmount": 50,
        "productCategory": "Electronics",
        "productImage": "https://i.pining.com/1200x/3f/8b/d7/3f8bd793fc081220462e80dee8fed87d.jpg",
        "productBrand": "CoolMe",
        "productDetail": "Compact mini desk fan powered via USB port. Features 3 adjustable speed settings with an ultra-quiet motor. Modern, space-saving design ideal for office
            desks, bedside tables, or traveling."
    },
    {
        "productID": "P00007",
        "adminID": "A007",
        "productName": "Plastic Food Container Set",
        "productAmount": 200,
        "productCategory": "Kitchen",
        "productImage": "https://i.pinimg.com/736x/62/92/3c/62923cfba40e4646839966088a1316d0.jpg",
        "productBrand": "SafeLock",
        "productDetail": "Set of 2 airtight food storage containers. Secure locking lids prevent leaks and keep food fresh for longer. Microwave safe (excluding lids) and easy to
            clean. Stackable design saves cabinet space."
    },
    {
        "productID": "P00008",
        "adminID": "A008",
        "productName": "Lifeventure USB Travel Adapter",
        "productAmount": 80,
        "productCategory": "Electronics",
        "productImage": "https://i.pining.com/1200x/25/42/47/2542474cff6ccbb1f7f0fefa95e0003f.jpg",
        "productBrand": "PowerUp",
        "productDetail": "Universal travel adapter compatible with outlets in over 160 countries. Includes dual USB ports for simultaneous charging of smartphones and tablets. Made
            from flame-retardant materials for extra safety."
    },
    {
        "productID": "P00009",
        "adminID": "A009",
        "productName": "Scented Candle - Jasmine",
        "productAmount": 120,
        "productCategory": "Home",
        "productImage": "https://i.pining.com/736x/d0/93/78/d09378d019c2981cfe21da16ab0ead86.jpg",
        "productBrand": "ThaiScent",
        "productDetail": "Natural jasmine-scented candle crafted to create a relaxing home atmosphere. Reduces stress with a long-lasting, soothing aroma. Low-smoke wick ensures clean
            burning. Perfect for bedrooms or living areas."
    },
    {
        "productID": "P00010",
        "adminID": "A010",
        "productName": "Pack of 5 Bamboo Chopsticks",
        "productAmount": 300,
        "productCategory": "Kitchen",
        "productImage": "https://m.media-amazon.com/images/I/61jEYJi9xML._AC_UF894,1000_QL80_.jpg",
        "productBrand": "Panda",
        "productDetail": "5 pairs of polished natural bamboo chopsticks. Splinter-free and smooth finish for a comfortable dining experience. Reusable and eco-friendly. Standard 24cm
            length suitable for all types of Asian cuisine."
    },

    {
        "productID": "P00011",
        "adminID": "A001",
        "productName": "Friction Powered Truck Toy",
        "productAmount": 108,
        "productCategory": "Toys",
        "productImage": "https://m.media-amazon.com/images/I/51sFqCGMt+L._SS400_.jpg",
        "productBrand": "MOZXEBOK",
        "productDetail": "Durable friction-powered truck toy. Simply push forward and watch it go! Made from high-quality ABS plastic, impact-resistant and safe for kids. Encourages
            motor skill development in children aged 3+."
    },
    {
        "productID": "P00012",
        "adminID": "A002",
        "productName": "Double Layer Race Truck",
        "productAmount": 124,
        "productCategory": "Toys",
        "productImage": "https://m.media-amazon.com/images/I/51zHaEO4n7L._SS400_.jpg",
        "productBrand": "Bstoyder",
        "productDetail": "Large double-layer race car hauler. Includes 4 small racing cars. Features realistic engine sounds and light effects. The back ramp opens to launch cars. A
            perfect gift for boys and girls who love vehicles."
    },
    {
        "productID": "P00013",
        "adminID": "A003",
        "productName": "Manual Coffee Grinder",
        "productAmount": 45,
        "productCategory": "Kitchen",
        "productImage": "https://image.makewebeasy.net/makeweb/m_1920x0/aSAKunEHs/etc/12_1.jpg",
        "productBrand": "Hillkoff",
        "productDetail": "Premium manual coffee bean grinder with adjustable grind settings. High-quality ceramic burr ensures a consistent grind without heat buildup, preserving the
            natural aroma and flavor of your coffee beans."
    },
    {
        "productID": "P00014",
        "adminID": "A004",
        "productName": "Desktop Organizer Box",
        "productAmount": 188,
        "productCategory": "Office",
        "productImage": "https://cdn01.pinkoi.com/product/UuRsYshh/7/640x530.jpg",
        "productBrand": "-",
        "productDetail": "Minimalist desktop organizer with multiple compartments for pens, paper clips, and sticky notes. Translucent matte finish fits any desk aesthetic. Keeps your
            workspace tidy and essential items within reach."
    },
    {
        "productID": "P00015",
        "adminID": "A005",
        "productName": "Yoga Mat Cleaning Spray",
        "productAmount": 99,
        "productCategory": "Health",
        "productImage": "https://m.media-amazon.com/images/I/818-3QRf+EL.jpg",
        "productBrand": "Asutra",
        "productDetail": "Organic yoga mat cleaner that eliminates 99.9% of bacteria and sweat odors. Infused with peppermint essential oil for a refreshing scent. Fast-drying,
            non-greasy formula safe for all mat materials."
    },
```

```
    {
        "productID": "P00016",
        "adminID": "A004",
        "productName": "LED Reading Light (Clip-on)",
        "productAmount": 75,
        "productCategory": "Electronics",
        "productImage": "https://m.media-amazon.com/images/I/7103tFwgMoL.jpg",
        "productBrand": "KTEBO",
        "productDetail": "Flexible clip-on reading light with a 360-degree gooseneck. Warm white LED light is easy on the eyes. Rechargeable via USB with up to 5 hours of continuous
            use. Ideal for late-night reading or studying."
    },
    {
        "productID": "P00017",
        "adminID": "A007",
        "productName": "Ceramic Ramen Bowl",
        "productAmount": 60,
        "productCategory": "Kitchen",
        "productImage": "https://images.squarespace-cdn.com/content/v1/60b65cba9c76c2177da28451/71c2b2e4-d86c-4a1a-a29c-49fd003fb415/IMG_6526.jpg",
        "productBrand": "Daphnia Ceramics",
        "productDetail": "Japanese-style ceramic bowl, extra-large 8-inch size. Heat resistant and durable. Hand-painted elegant patterns make it a beautiful addition to your kitchen.
            Perfect for ramen, noodles, or hearty soups."
    },
    {
        "productID": "P00018",
        "adminID": "A008",
        "productName": "Universal Screen Protector Kit",
        "productAmount": 200,
        "productCategory": "Electronics",
        "productImage": "https://i5.walmartimages.com/seo/WriteRight-Universal-Screen-Protector-for-Screens-up-to-6-8-2-Pack_1c3773e2-b5f2-41bd-856a-3280a185698d.
            d98b9c1a483e4ae9f88ee7f30dec01b1.jpeg",
        "productBrand": "-",
        "productDetail": "Complete screen protection kit including universal film, screen cleaning solution, and a lint-free cloth. Prevents scratches, smudges, and fingerprints.
            Compatible with most smartphones and tablets."
    },
    {
        "productID": "P00019",
        "adminID": "A009",
        "productName": "Essential Oil Diffuser",
        "productAmount": 35,
        "productCategory": "Home",
        "productImage": "https://www.innogear.com/cdn/shop/files/
            InnoGearEssentialOilDiffuser_150MLCeramicDiffuserForHomeHandcraftedAromatherapyDiffuserUltrasonicCoolMistHumidifierwith2MistModesWaterlessAuto0ffforRoomOffice_Black_7_2480x.
            jpg?v=1703586449",
        "productBrand": "InnoGear",
        "productDetail": "Portable ultrasonic humidifier and essential oil diffuser. Enhances air moisture while dispersing soothing scents. Features 7-color changing LED lights.
            Auto-shutoff safety feature when water runs out."
    },
    {
        "productID": "P00020",
        "adminID": "A010",
        "productName": "Gardening Tool Set (3 Pcs)",
        "productAmount": 110,
        "productCategory": "Garden",
        "productImage": "https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQeqOyZ_6QbaE-grQjsnzVVoeJXpPFPuio9wg&s",
        "productBrand": "GreenThumb",
        "productDetail": "Mini 3-piece gardening set including a trowel, transplanter, and rake. Made from rust-resistant steel with ergonomic wooden handles. Perfect for indoor
            houseplants, succulents, or balcony gardening."
    }
  ]
}
```


Test Case 2: Search with multiple criteria
Method: Get
URL: http://localhost:3100/products/search?name=Pen&brand=Double&category=Stationery
Result:


```
{
    "error": 0,
    "count": 1,
    "data": [
        {
            "productID": "P00002",
            "adminID": "A002",
            "productName": "Pack of 3 Blue Ballpoint Pens",
            "productAmount": 300,
            "productCategory": "Stationery",
            "productImage": "https://pim-cdn0.ofm.co.th/products/original/1091584.jpg?v=20260330&x-image-process=image/format,webp",
            "productBrand": "Double A",
            "productDetail": "Set of 3 premium blue ink ballpoint pens with a 0.5mm needle tip. Provides a smooth, skip-free writing experience. Quick-drying ink prevents smudging, making
                it ideal for students and office professionals."
        }
    ]
}
```


Test Case 3: Get product detail
Method: Get
URL: http://localhost:3100/products/P00002
Result:


```
{
    "error": 0,
    "data": {
        "productID": "P00002",
        "adminID": "A002",
        "productName": "Pack of 3 Blue Ballpoint Pens",
        "productAmount": 300,
        "productCategory": "Stationery",
        "productImage": "https://pim-cdn0.ofm.co.th/products/original/1091584.jpg?v=20260330&x-image-process=image/format,webp",
        "productBrand": "Double A",
        "productDetail": "Set of 3 premium blue ink ballpoint pens with a 0.5mm needle tip. Provides a smooth, skip-free writing experience. Quick-drying ink prevents smudging, making it
            ideal for students and office professionals."
    }
}
```

Test Case 4: Get product detail (Not Found)
Method: Get
URL: http://localhost:3100/products/P99999
Result:


```
{
    "error": 1,
    "message": "Product not found."
}
```

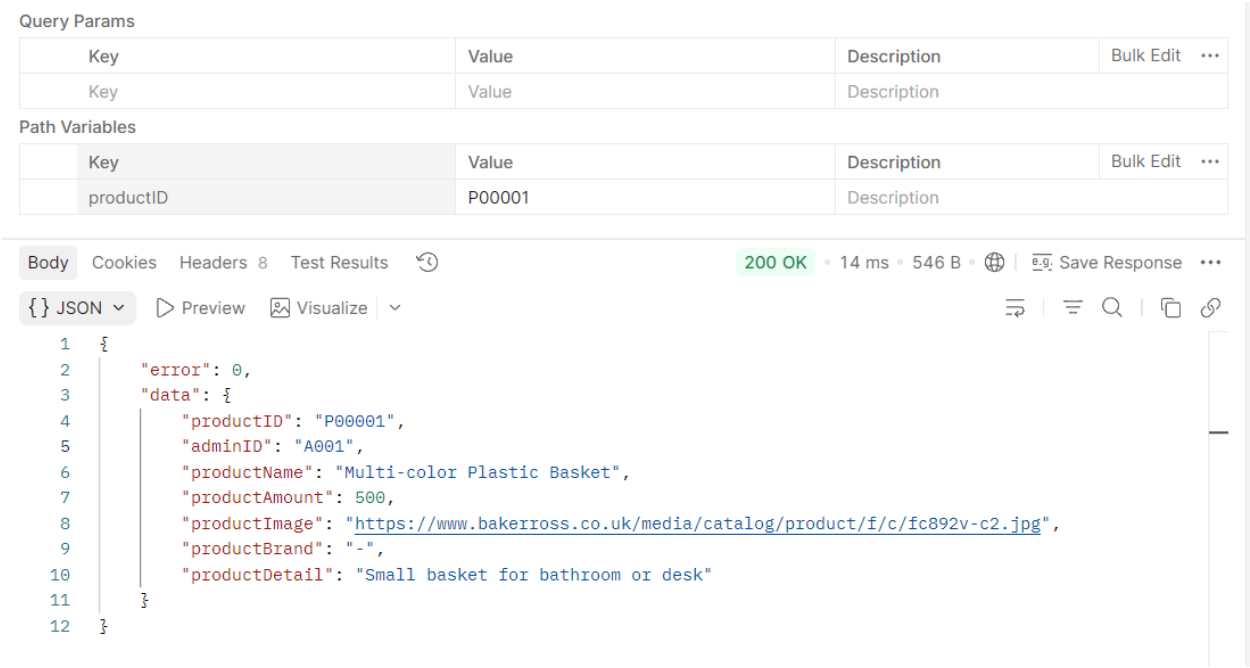

Test Case 5: GET all products
Headers: Authorization: supersecret
Method: Get
URL: http://localhost:3100/products/admin/
Result:


```
{
    "error": 0,
    "data": [
        {
            "productID": "P00001",
            "adminID": "A001",
            "productName": "Multi-color Plastic Basket",
            "productAmount": 500,
            "productImage": "https://www.bakerross.co.uk/media/catalog/product/f/c/fc892v-c2.jpg",
            "productBrand": "-",
            "productDetail": "Small basket for bathroom or desk"
        },
        {
            "productID": "P00002",
            "adminID": "A002",
            "productName": "Pack of 3 Blue Ballpoint Pens",
            "productAmount": 300,
            "productImage": "https://pim-cdn0.ofm.co.th/products/original/1091584.jpg?v=20260330&x-image-proces
            "productBrand": "Double A",
            "productDetail": "Smooth writing blue ink pens"
        },
        {
            "productID": "P00003",
            "adminID": "A003",
            "productName": "Stainless Steel Vegetable Peeler",
            "productAmount": 150,
            "productImage": "https://www.vipshopboutic.com/cdn/shop/products/178172_2.jpg?v=1692863993&width=1
            "productBrand": "vipshopboutic",
            "productDetail": "Sharp blade for easy peeling"
        },
```

```
    {
        "productID": "P00004",
        "adminID": "A004",
        "productName": "Microfiber Cleaning Cloth",
        "productAmount": 400,
        "productImage": "https://www.nitori.co.th/cdn/shop/files/898245132_640x640.jpg?v=1730701389",
        "productBrand": "NITORI",
        "productDetail": "High absorbency for kitchen use"
    },
    {
        "productID": "P00005",
        "adminID": "A005",
        "productName": "Crispy Fish Snack Pack",
        "productAmount": 250,
        "productImage": "https://st.bigc-cs.com/cdn-cgi/image/format=webp,quality=90/public/media/catalo
        "productBrand": "Taro",
        "productDetail": "Small snack pack, spicy flavor"
    },
    {
        "productID": "P00006",
        "adminID": "A006",
        "productName": "Gaiatop Mini Desk Fan (USB)",
        "productAmount": 50,
        "productImage": "https://i.pinimg.com/1200x/3f/0b/d7/3f0bd793fc001220462e80dee8fed57d.jpg",
        "productBrand": "CoolMe",
        "productDetail": "Battery or USB powered desk fan"
    },
    {
        "productID": "P00007",
        "adminID": "A007",
        "productName": "Plastic Food Container Set",
        "productAmount": 200,
        "productImage": "https://i.pinimg.com/736x/62/92/3c/62923cfba40e4646839966005a1316d0.jpg",
        "productBrand": "SafeLock",
        "productDetail": "Set of 2 stackable plastic boxes"
    },
    {
        "productID": "P00008",
        "adminID": "A008",
        "productName": "Lifeventure USB Travel Adapter Plug",
        "productAmount": 80,
        "productImage": "https://i.pinimg.com/1200x/25/42/47/2542474cff6ccbb1f7f0fefa95e0003f.jpg",
        "productBrand": "PowerUp",
        "productDetail": "Compact adapter for all outlets"
    },
    {
        "productID": "P00009",
        "adminID": "A009",
        "productName": "Scented Candle - Jasmine",
        "productAmount": 120,
        "productImage": "https://i.pinimg.com/736x/d0/93/78/d09378d019c2901cfe21da16ab0ead06.jpg",
        "productBrand": "ThaiScent",
        "productDetail": "Standard sized 20-Baht candle"
    },
```

```
        {
            "productID": "P00010",
            "adminID": "A010",
            "productName": "Pack of 5 Bamboo Chopsticks",
            "productAmount": 300,
            "productImage": "https://m.media-amazon.com/images/I/61jEYji9xML._AC_UF894,1000_QL80_.jpg",
            "productBrand": "Panda",
            "productDetail": "Natural bamboo chopsticks"
        }
    ]
}
```


Test Case 6: GET product by ID
Headers: Authorization: supersecret
Method: GET
URL: http://localhost:3100/products/admin/:productID



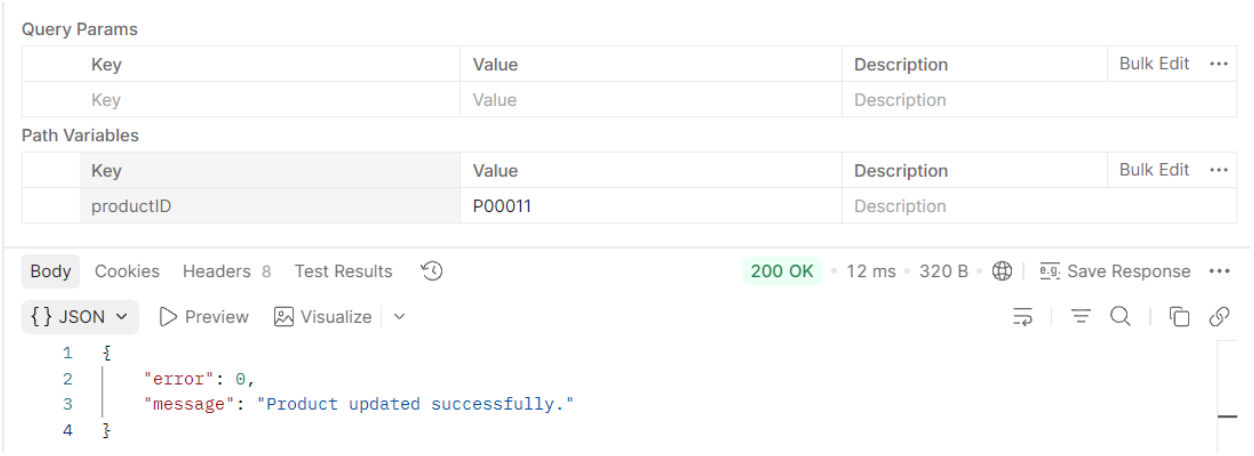
Test Case 7: POST create new product
Headers: Authorization: supersecret
Method: POST
URL: http://localhost:3100/products/admin/
Body: raw JSON

```
{
  "productID": "P00011",
  "adminID": "A001",
  "productName": "Test Product",
  "productAmount": 100,
  "productImage": "https://example.com/image.jpg",
  "productBrand": "BrandX",
  "productDetail": "Some detail"
}
```

```
{
    "error": 0,
    "message": "Product created successfully.",
    "productID": "P00011"
}
```


Test Case 8: PUT update product
Headers: Authorization: supersecret
Method: PUT
URL: http://localhost:3100/products/admin/:productID
Body: raw JSON


```
{
  "productName": "Updated Name",
  "productAmount": 200,
  "productImage": "https://example.com/newimage.jpg",
  "productBrand": "NewBrand",
  "productDetail": "Updated detail"
}
```

Test Case 9: DELETE product
Headers: Authorization: supersecret
Method: DELETE
URL: http://localhost:3100/products/admin/:productID

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Path Variables

Key	Value	Description	Bulk Edit
productID	P00011	Description	

Body Cookies Headers 8 Test Results 200 OK 13 ms 320 B Save Response

{ } JSON Preview Visualize


```
{
    "error": 0,
    "message": "Product deleted successfully."
}
```